VALUATION SCHEME

QPCODE: 24900200

MAHATMAGANDHIUNIVERSITY, KOTTAYAM

FIRST SEMESTERMGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2024

First Semester

Discipline Specific Core Course - MG1DSCPEG100 – FUNDAMENTALS OF PETROLEUM GEOCHEMISTRY

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours Maximum Marks: 50

Part A

Multiple Choice Questions Answer All Questions.

Each question carries one mark. (10 × 1= 10)

1. Petroleum is primarily composed of which type of hydrocarbons? [K][1]
   D, All of the above
2. The Abiogenic Theory posits that petroleum originates from: [U][1]
   C, Deep Earth processes involving Carbon and hydrogen compounds
3. Which factor does NOT directly influence the migration of petroleum? [K][1]
   C, Biological activity
4. Recognize the key benefit of using remote sensing in petroleum exploration [K][3]
   C, Wide area coverage to identify surface anomalies
5. Select the well-logging technique measures the natural radioactivity of rock formations [K][3]
   C, Gamma-Ray Logging
6. Recognize the primary purpose of conducting a resistivity log in a borehole? [K][3]
   B, To detect the presence of hydrocarbons by measuring the resistivity of rock formations
7. Select which of the following hydrocarbon is primarily found in natural gas? [K][4]
   C, Methane
8. Identify the role of sulphur compounds in crude oil refining [U][4]
   C, They can lead to corrosion
9. Recognize which hydrocarbon is typically liquid at room temperature? [K][5]
   C, Hexane
10. Identify which of the following hydrocarbons would generally have the highest flash point? [K][5]
    C, Diesel

Part B

Short Answer Type Questions Answer any four questions.

Each question carries three marks. (4 × 3= 12)

(Give Full credit to any 3 Points of corresponds to each questions)

11. List out the benefits of deasphalting process [An] [1]
    - Decreases asphaltane, resin content, metal amounts, carbon residue, aromatic content, nitrogen and sulphur content
    - Decreases viscosity, Increases pour point and API gravity.

12. Illustrate migration of hydrocarbons from carrier beds to reservoir rock [A] [2]
    - The movement of petroleum from the source rock (carrier beds) to the reservoir rock is called 2° migration.
    - During migration composition of petroleum may differ due to filtration, adsorption, and reaction with elements like sulphur.
    - The cap rocks of the basement rock should have low permeability than the reservoir rock to prevent the escape of oil and gas.
13. Write a short note on magnetic method in petroleum exploration and its key applications [A] [3]
    - Geophysical exploration technique used to map variations in the earth's magnetic field caused by the presence of different types of rocks.
    - Rocks containing iron or magnetite, have distinct magnetic properties, and these variations can provide clues about subsurface geological structures that may trap hydrocarbons.
    - The magnetic method is primarily used to:Map basement structures & Identify faults and fractures .
14. Describe the role of trenching and pitting in petroleum exploration. [U] [3]
    - Trenching involves digging long, narrow ditches (trenches) in the ground to expose bedrock or other subsurface materials.
    - Pitting is a similar technique, but instead of a long trench, pits are excavated in the ground.
    - Pitting is often used in combination with geochemical exploration to analyze soil or rock samples for traces of hydrocarbons or other mineral anomalies.
15. Summarize the implications of nitrogen compounds for crude oil pricing? [U] [4]
    - The presence of nitrogen compounds in crude oil can significantly impact pricing due to their influence on refining efficiency and product yield.
    - High nitrogen content is often associated with heavier, lower-quality crudes, which may require more extensive and costly processing.
    - Lighter crudes with minimal nitrogen content are favored for their easier processing and higher yields of valuable products.
16. Explain the significance of a high calorific value for a fuel [U] [5]
    - Calorific value is the amount of energy released when a unit mass or volume of fuel is completely burned, typically expressed in mega joules per kilogram (MJ/kg) or mega joules per cubic meter (MJ/m^3).
    - It is a critical measure for evaluating the energy content of fuels, influencing their efficiency and suitability for various applications.
    - Higher calorific values indicate more energy-efficient fuels, making them preferable for combustion processes in power generation, transportation, and heating.

Part C

Short Essay Type Questions

Answer any four questions. Each question carries seven marks. ($7 \times 4 = 28$)

17. Justify that modern theory provides better explanation for origin of petroleum than other

theories. [E] [1]

(Description of Modern Theory 2 Marks, Justification 4 Marks)

- The modern theory for the origin of petroleum, known as the biogenic theory, posits that petroleum is primarily derived from the decay of ancient marine organism such as plankton and algae, which were buried under layers of sediment millions of years ago, undergo anaerobic decomposition by bacteria in which the fat content from the sediments get converted into fatty acids. These fatty acids undergo further transformation to form kerogen. Due to combined action of heat and pressure, the saprophytic kerogen converted into oil and gas. This oil and gas undergo 1° migration from fine grained rocks to coarse grained sedimentaryrocks. 2° migration of this fluids leads to accumulation which result in formation of oil wells.

Key reasons that make it a more compelling explanation for the origin of petroleum compared to alternative theories

- The biogenic theory is supported by extensive chemical and geological data.
- Fossils of ancient marine microorganisms are found in petroleum-rich formations.
- Laboratory experiments also supports the theory
- Sedimentary structures like hydrocarbon traps and reservoirs are as easily explained by the modern -biogenic theory

18. What is primary migration in the context of petroleum formation, and how does it differ from secondary migration? [K] [2]

(Explanation on primary migration 3.5 marks, secondary migration 3.5 marks)

- 1° migration: Migration of petroleum starts in the source rocks. Primary migration refers to the initial movement of hydrocarbons from the source rock, where they are generated, to the adjacent carrier beds. Primary migration is closely tied to the generation of hydrocarbons.
- 2° migration: The movement of petroleum from the source rock to the reservoir rock is called 2° migration, often facilitated by geological structures like faults and fractures. This focuses on the accumulation and trapping of these hydrocarbons in suitable reservoir rocks. During migration composition of petroleum may differ due to filtration, adsorption, and reaction with elements like sulphur. The cap rocks of the basement rock should have low permeability than the reservoir rock to prevent the escape of oil and gas.

19. Illustrate how core sampling enhances the understanding of subsurface geology. Discuss its application in oil exploration [An] [3]

(Explanation on Core sampling 3.5 marks, Application 3.5 marks)

- Core sampling is a direct method of obtaining a cylindrical section of rock from a wellbore during drilling.A special tool called a core barrel is lowered into the borehole and used to cut and extract a continuous section of the rock, known as a core.

  Application in petroleum exploration:
- Lithology and stratigraphy: Porosity and permeability: Fluid content and saturation: Rock mechanics: While core sampling is more expensive and time-consuming than other exploration techniques, it provides a wealth of detailed information that is crucial for accurate reservoir characterization and decision-making in oilfield development.

20. Discuss the various surface and subsurface exploration techniques used in petroleum

exploration.[U][3]

(Any 7 exploration method give Full Credit, - 1 Mark each - At least 3 method from surface or subsurface exploration techniques method)

Surface Exploration Techniques:

- Geological Exploration: Photogeology: Remote Sensing: Geographic Information Systems (GIS): Geochemical Exploration: Geobotanical Exploration:

Subsurface Exploration Techniques:

- Gravity Method: Magnetic Method: Electrical Methods (Self-Potential, Resistivity): Electromagnetic Methods: Seismic Methods: Borehole Logging:.

21. Explain the significance of metallic and organo metallic hydrocarbons in crude oil and their implications for refining. [U] [4]

(Any three points on significance of metallic constituent 3.5 marks, organo metallic hydrocarbon 3.5 marks)

- Metallic constituents, such as nickel, vanadium, and iron, are often found in crude oil and can have profound effects on refining efficiency and product quality. The presence of these metals can also cause fouling in equipment, resulting in increased maintenance costs and operational downtime, cause corrosion of refinery equipment.
- Organometallic hydrocarbons, which contain metal atoms bonded to organic molecules, can be significant in crude oil and its refining processes. The presence of organometallic compounds can complicate the behavior of catalysts and may lead to unwanted side reactions, resulting in decreased efficiency and product yield.

22. Explain how the refractive index is used to characterize hydrocarbons and assess their quality.[U][5] **(Give full credit to 7 points)**
    - The refractive index of a hydrocarbon is a measure of how much light bends as it passes through the substance.
    - It is defined as the ratio of the speed of light in a vacuum to the speed of light in the medium.
    - It provides valuable information about their composition, purity, and concentration.
    - A higher refractive index often correlates with a higher concentration of aromatic compounds
    - A higher refractive index can impact the performance and characteristics of the final product.
    - Refractive index measurements can assist in identifying different types of hydrocarbons, such as alkanes, alkenes, and aromatics.
    - By monitoring the refractive index, manufacturers can ensure compliance with industry standards and maintain product consistency, ultimately enhancing operational efficiency and safety.
